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Stochastic ALNS for the stochastic team orienteering problem

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HANDBOOK

OPERATIONAL RESEARCH SOCIETY

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solutions and in particular examine how cooperation among multiple companies can overcome these challenges. By employing game theoretic tools, we aim to provide insights into the dynamic relationship between collaborative energy production, storage and consumption in off-grid systems. In particular, we study how economic incentives should be designed to overcome hesitance of companies to collaborate.

■ MC-20

Monday, 12:30-14:00 - Room: Esther Simpson 2.11

Metaheuristic algorithms

Stream: Combinatorial Optimization
Invited session
Chair: *David Pisinger*

1 - Random-Key Optimizers for Solving Combinatorial Optimization Problems
Antonio Chaves, Mauricio Resende

This work presents the Random-Key Optimizer (RKO), a modular stochastic local search framework for combinatorial optimization. RKO encodes solutions as random-key vectors that are transformed into feasible solutions using problem-specific decoders. The framework integrates multiple metaheuristics: simulated annealing, iterated local search, variable neighborhood search, particle swarm optimization, genetic algorithms, large neighborhood search, and greedy randomized adaptive search procedures. These metaheuristics can operate independently or collaboratively through an elite solution pool. RKO's problem-independent architecture requires only a decoder implementation to adapt to new problems. The framework features novel continuous-space local search heuristics using randomized variable neighborhood descent that combines an adapted Nelder-Mead with other optimization techniques. Implemented in C++ with multithreading capabilities, RKO enables solution sharing among concurrent metaheuristics. We demonstrate its effectiveness on four NP-hard problems: the tree of hubs location problem, traveling thief problem, portfolio problem, and node-capacitated graph partitioning problem. Results confirm RKO's robustness and adaptability as an effective tool for combinatorial optimization. The RKO solver is publicly available on GitHub.

2 - A scenario decomposition metaheuristic for solving two-stage stochastic problems with integrality constraints
Kristine Børsting, David Pisinger

Many important modern-day problems involve uncertainty, such as meeting electricity demand with a growing fraction of variable energy sources. Stochastic programs facilities making decisions based on all realizations of the uncertain parameters, however the number of variables in these models grow with the number of scenarios considered. Several years of historical data may be available, providing a rich description of the uncertainty, but if the problem contains integrality constraints it is limited how many scenarios can be considered before it becomes intractable.

Progressive Hedging is a scenario decomposition method originally developed for convex problems. The non-anticipativity constraints are Lagrangian relaxed, making it possible to solve a series of computationally easier subproblems that only account for a few scenarios each. Consensus is then enforced between subproblems through objective penalties. This method has also been used as a heuristic for problems with integer variables, however, only minor modifications have been made to the framework which still reflects a non-integer design.

In this presentation, we will introduce novel improvements to the Progressive Hedging framework for use as a metaheuristic for stochastic linear problems with integer variables. Computational results for various two-stage stochastic optimization problems underline its general usefulness.

3 - Enhancing the convergence of the Feasibility Pump through variable decomposition
Lucas Assuncao, Sebastian Urrutia, Andréa Cynthia Santos

Feasibility Pump (FP) is a well-known heuristic for mixed integer linear programming that iteratively moves fractional solutions toward integer feasibility by minimizing their distance to target integer points while maintaining the original linear constraints. Due to its simplicity and effectiveness, FP is widely used in commercial solvers, with many variations improving its convergence and solution quality. This work introduces a novel acceleration technique relying on the intuition that not all binary variables are active in a feasible solution. A kernel search strategy identifies a subset of promising binary variables, while the remaining ones are grouped into ranked buckets based on their likelihood of appearing in a feasible solution. FP sub-problems prioritize the kernel and gradually incorporate other variables as needed. To our knowledge, this is the first decomposition-based approach within the FP framework. Computational experiments on MIPLIB 2017 benchmark and two additional problems in vehicle routing and scheduling show improved success rates and reduced running times. Notably, the proposed method surpasses FP's performance and finds feasible solutions in cases where CPLEX fails within an hour.

4 - Stochastic ALNS for the the stochastic team orienteering problem
David Pisinger

The team orienteering problem is a generalization of the selective traveling salesman problem. A fleet of vehicles is available to visit a subset of customers within a given time limit. Each customer has an associated profit, and the profit can only be collected once. In the stochastic version of the problem, the customers are split into subscription customers that must be visited every day, and stochastic on-demand customers. The task is to select a subset of the subscription customers, such that the expected profit of served subscription and on-demand customers is maximized.

We present a stochastic Adaptive Large Neighborhood Search (SALNS) algorithm for the problem, in which an upper-level local search method operates on the first-stage decision variables, while a number of lower-level local search methods optimize the individual scenarios. Both customers, demands and travel times can be stochastic as long as they can be represented by a number of scenarios. Various destroy and repair neighborhoods are presented for both the upper-level and lower-level ALNS algorithms.

Computational experiments show that the algorithm scales well with the number of scenarios, making it possible to solve instances with up to 1000 customers and 64 scenarios.

■ MC-21

Monday, 12:30-14:00 - Room: Esther Simpson 2.12

Irregular packing and cutting

Stream: Cutting and packing (ESICUP)
Invited session
Chair: *Tetyana Romanova*

1 - Algorithms for Bar Nesting Software
Rhyd Lewis

In manufacturing and construction industries, bar nesting software is often used to help optimise the cutting patterns of linear materials such as reinforcing bars, girders, pipes, wooden joists, and window frames. The primary objective of such software is to minimise material wastage by determining efficient arrangements for cutting the desired items from lengthy stock materials. This process not only reduces costs but also supports sustainable practices by maximizing resource utilization. Several companies offer bar nesting in construction-focused CAD tools, including AutoRebar, RebarCAD, AutoBarSizer, and TopSolid Design.